

Experiment 3: Information Theory

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Posterior Probabilities: [[0.0519112 0.15000000]]
(base) bhanu@Bhanus-MacBook-Air C01-ASS3 % python Q3.py
Entropy of Dataset = 0.9403

Information Gain
Outlook : 0.2467
Temperature : 0.0292
Humidity : 0.1518
Wind : 0.0481

Best Attribute = Outlook
(base) bhanu@Bhanus-MacBook-Air C01-ASS3 %
```

Result

The Play Tennis dataset was successfully analyzed using Information Theory. The entropy of the dataset was calculated as **0.9403**. Information Gain was computed for all input attributes.

The calculated Information Gain values were:

- Outlook = 0.2467
- Temperature = 0.0292
- Humidity = 0.1518
- Wind = 0.0481

Among all attributes, **Outlook** produced the highest Information Gain. Therefore, Outlook was selected as the best attribute for the first split in the Decision Tree.